XRPL Validator — Daily Engineering Report · 2026-09-29

Period: 2026-09-29 00:00 to 24:00 US Eastern (2026-09-29 04:00 to 2026-09-30 04:00 UTC); ledgers #107,306,985 to #107,329,221. Every count from the ledger and the validator's logs stops at #107,329,221. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,228 of the 22,237 ledgers the network closed (99.960%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 9 during planned restarts.
- **Native engine:** 2,305 engine receipts recorded (2305 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 1 deployment (cycle 161), 3 commits merged into main, 2 engine findings fixed (F407–F408).

Service levels

Measure	Value
Ledgers closed by the network	22,237
Validations issued	22,228
Coverage (validations / ledgers closed)	99.960%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	9
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	1
Basis: the hash the first peer announced as accepted (before cycle 183)	22,228

Restarts:

- 14:57:59 UTC: deployment of cycle 161; validations resumed at #107,317,149 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,306,985–#107,329,221), by step:

Step	Ledgers with a receipt
Step 1 · native engine	2,305
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,311,328	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 2E1ABE61 (deleted), E9616FF7, FEDA226B; 2 object(s) differed in content	fixed: F407, commit e1140ef, deployed in cycle 161 (2026-09-29 14:58 UTC)
R-02	—	#107,314,214	Step 1 · native engine	2 object(s) differed in content	fixed: F408, commit 1fd1b9b, deployed in cycle 161 (2026-09-29 14:58 UTC)
R-03	—	#107,314,806	Step 1 · native engine	318 change(s) of ours the network did not make	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)

ID	Time	Ledger(s)	Step	Description	Disposition
R-04	—	#107,314,807	Step 1 · native engine	22 transaction result(s) differed from rippled's; 50 ledger change(s) the network made were absent from ours: 02BAAC1E, 7B73A610, F0B9A528; 4 change(s) of ours the network did not make; 18 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-05	—	#107,314,808	Step 1 · native engine	6 transaction result(s) differed from rippled's; 25 ledger change(s) the network made were absent from ours: 368C81F8, E94EC39E, 2DAC0D23; 10 change(s) of ours the network did not make; 6 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-06	—	#107,314,809	Step 1 · native engine	18 transaction result(s) differed from rippled's; 35 ledger change(s) the network made were absent from ours: 7F0234EF, 36E4369F, BCA8DC84; 9 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-07	—	#107,314,810	Step 1 · native engine	13 transaction result(s) differed from rippled's; 25 ledger change(s) the network made were absent from ours: E5DE2D20, D0C02E3C, 9703D8A9; 9 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-08	—	#107,314,812	Step 1 · native engine	3 transaction result(s) differed from rippled's; 12 ledger change(s) the network made were absent from ours: 41DD81B6, D9AEEB7F, 19DA952A; 5 change(s) of ours the network did not make; 5 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-09	—	#107,314,813	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 152E488E, 6533684B, 32E7800A; 4 change(s) of ours the network did not make; 13 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-10	—	#107,314,814	Step 1 · native engine	2 transaction result(s) differed from rippled's; 6 ledger change(s) the network made were absent from ours: B335DFD1, 280D8E8A, 5E4E9106 (deleted); 4 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-11	—	#107,314,815	Step 1 · native engine	10 change(s) of ours the network did not make; 8 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-12	—	#107,314,816	Step 1 · native engine	1 transaction result(s) differed from rippled's; 8 change(s) of ours the network did not make; 7 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-13	—	#107,314,817	Step 1 · native engine	8 change(s) of ours the network did not make; 4 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state

ID	Time	Ledger(s)	Step	Description	Disposition
					hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-14	—	#107,314,818	Step 1 · native engine	5 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-15	—	#107,314,819	Step 1 · native engine	7 ledger change(s) the network made were absent from ours: 7ED917AB, B1B4255A, F98ACA7B; 3 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-16	—	#107,314,820	Step 1 · native engine	1 transaction result(s) differed from rippled's; 5 ledger change(s) the network made were absent from ours: 665A4EBE, E7C0008E, E5BF1B3C (deleted); 7 change(s) of ours the network did not make; 8 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-17	—	#107,314,821	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: B335DFD1; 2 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-18	—	#107,314,822	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 ledger change(s) the network made were absent from ours: D15897A1, 39AA5E1A, 860F5FD1 (deleted); 11 change(s) of ours the network did not make; 11 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-19	—	#107,314,823	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: 310C9855, D15897A1, 61FF29AB (deleted); 6 change(s) of ours the network did not make; 6 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-20	—	#107,314,824	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: D15897A1; 1 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-21	—	#107,314,827	Step 1 · native engine	2 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: 29019262, C6F1E946; 4 change(s) of ours the network did not make; 4 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-22	—	#107,314,828	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: 063BC4DA, 19DA952A; 5 change(s) of ours the network did not make; 4 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-23	—	#107,314,830	Step 1 · native engine	2 transaction result(s) differed from rippled's; 14 change(s) of ours the network did not make; 5 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at

ID	Time	Ledger(s)	Step	Description	Disposition
					14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-24	—	#107,314,831	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 96556427, 7CE01B93, CFEC8953; 4 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-25	—	#107,314,832	Step 1 · native engine	2 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-26	—	#107,314,837	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: AF4C3C6B, 06FC0CDE, AB4F967F; 1 change(s) of ours the network did not make; 6 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-27	—	#107,314,838	Step 1 · native engine	5 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: ACF13679, 68620CD2; 10 change(s) of ours the network did not make; 18 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-28	—	#107,314,839	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-29	—	#107,314,840	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: BA622EBB (deleted), 310C9855, 3DE9599C (deleted); 1 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-30	—	#107,314,841	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 39AA5E1A, 860F5FD1 (deleted), A6BD8AB0 (deleted); 2 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-31	—	#107,314,842	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: 18133E5D, 42821437, 19DA952A; 8 change(s) of ours the network did not make; 4 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-32	—	#107,314,846	Step 1 · native engine	9 change(s) of ours the network did not make; 6 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-33	—	#107,314,847	Step 1 · native	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay,

ID	Time	Ledger(s)	Step	Description	Disposition
			engine	make; 4 object(s) differed in content	so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-34	—	#107,314,848	Step 1 · native engine	2 transaction result(s) differed from rippled's; 8 ledger change(s) the network made were absent from ours: 1A4447F8, 97D1320C, 318E00F1; 9 change(s) of ours the network did not make; 3 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-35	—	#107,314,849	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 4A7F81F3, B1B4255A, 6A87108A; 1 change(s) of ours the network did not make; 3 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-36	—	#107,314,850	Step 1 · native engine	6 change(s) of ours the network did not make; 3 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-37	—	#107,314,851	Step 1 · native engine	8 change(s) of ours the network did not make; 3 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-38	—	#107,314,852	Step 1 · native engine	7 change(s) of ours the network did not make; 4 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-39	—	#107,314,854	Step 1 · native engine	1 transaction result(s) differed from rippled's; 8 change(s) of ours the network did not make; 5 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)
R-40	—	#107,314,857	Step 1 · native engine	3 transaction result(s) differed from rippled's; 3 ledger change(s) the network made were absent from ours: 54123B87 (deleted), 4A7F81F3 (deleted), C6F1E946; 15 change(s) of ours the network did not make; 5 object(s) differed in content	fixed: the native shadow's mirror lost ledger #107,314,806 to an unverified, empty FFI overlay, so every later ledger differed until the restart at 14:57:59 UTC rebuilt it; the validator's own state hash matched throughout (fix 50a05ae); deployed in cycle 161 (2026-09-29 14:58 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-29-01, INC-01 is INC-2026-09-29-01. Times are not shown where the reference node no longer holds the ledger.

2,265 more receipts (#107,314,858–#107,317,138) are in the record and not listed individually; their dispositions: 2,265 fixed.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
161	`50a05ae`	14:58:47 UTC	1,697 passed / 0 failed; port 745 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	411 matched, 0 mismatched	#36 from 15:25 UTC: receipts=1 (2026-09-30 15:32 UTC)

Cycle 161 delivered: `e1140ef` fix(xrpl-ledger): F407 — a stored QualityIn of QUALITY_ONE is no interest; the default line is deleted; `1fd1b9b` fix(xrpl-ledger): F408 — an AMM deposit mints the tokens its mode computes; LPTokenOut is never the mint; `50a05ae` fix(xrpl-node): the era sentinel walks the AccountRoot's fields; an unverified ledger re-hydrates the native shadow

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
09:06 UTC	xrpl-validator	`e1140ef`	fix(xrpl-ledger): F407 — a stored QualityIn of QUALITY_ONE is no interest; the default line is deleted
12:16 UTC	xrpl-validator	`1fd1b9b`	fix(xrpl-ledger): F408 — an AMM deposit mints the tokens its mode computes; LPTokenOut is never the mint
13:42 UTC	xrpl-validator	`50a05ae`	fix(xrpl-node): the era sentinel walks the AccountRoot's fields; an unverified ledger re-hydrates the native shadow

Findings fixed

- **fix(xrpl-ledger): F407 — a stored QualityIn of QUALITY_ONE is no interest; the default line is deleted** (`e1140ef` , xrpl-validator) Mainnet #107311328 59262A2D67A7: a TrustSet (limit 0, tfSetNoRipple + tfClearFreeze) on a CNY line whose holder's side stores HighQualityIn = 1,000,000,000 (QUALITY_ONE) and otherwise has no interest: limit 0, balance 0, NoRipple set on a holder without lsfDefaultRipple. rippled's SetTrust reads the line's stored qualities and zeroes QUALITY_ONE before the reserve decision (TrustSet.cpp `if (QUALITY_ONE == …
- **fix(xrpl-ledger): F408 — an AMM deposit mints the tokens its mode computes; LPTokenOut is never the mint** (`1fd1b9b` , xrpl-validator) Mainnet #107314214 5FC90948A0CD: a tfSingleAsset AMMDeposit of 1,021,192 drops into an XRP/FUZZY pool with LPTokenOut 56,639.2869999. In tfSingleAsset mode LPTokenOut is an optional MINIMUM: singleDeposit mints what the Amount buys (lpTokensOut, adjustLPTokensOut, adjustAssetInByTokens) and hands the field to deposit() as lpTokensDepositMin, which refuses tecAMM_FAILED when lpTokensDepositActual < …

Other engineering changes

- **fix(xrpl-node): the era sentinel walks the AccountRoot's fields; an unverified ledger re-hydrates the native shadow** (`50a05ae`, xrpl-validator) Mainnet #107314806 (soak #35, 08:26 EDT 2026-09-29): the era sentinel read the Sequence of the ledger's first seq-bearing transaction's sender (Payment C4F7C77C779F, Sequence 62,037,183) out of its AccountRoot with a byte search for 0x24. The account's Flags are 0x24000000 (lsfDisallowIncomingTrustline | lsfDisallowIncomingNFTokenOffer), so the search hit the Flags' first byte and read Sequence 36: a false …

Independence program

No independence record for UTC day 2026-09-29; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Soak #35: two engine receipts, both fixed; ended by the deployment below.** Soak #35 of build 0145757 began at 17:58 EDT on 2026-09-28 (seed ledger 107,300,970) and ended with the deployment of 50a05ae at 10:58 EDT, after about 17 hours. It recorded two engine receipts, F407 at 04:46 EDT and F408 at 07:54 EDT, both below.

Every check during it found no state-hash mismatch (the last at 09:17 EDT), with the validator on its primary reference node and no failovers: the receipts are differences in the native engine's own results, found by its comparison against libxrpl, not in the validator's state. From 08:26 EDT the comparison itself was out of step (below).

- **Finding F407 (a trust line whose only setting is the default quality).** At ledger 107,311,328 an account sent a TrustSet on a CNY trust line, setting the limit to 0, setting NoRipple and clearing freeze. Afterwards the line held no balance, the account's limit was 0, and NoRipple is that account's default (it does not enable DefaultRipple); the only other thing stored for its side was an incoming quality of 1,000,000,000. That value, QUALITY_ONE, is a rate of exactly 1, the same as no quality at all, and rippled's TrustSet treats a stored QUALITY_ONE as zero before deciding whether either side still has an interest in the line. Neither side did, so mainnet deleted the line: the line and the issuer's directory page went, the account's directory page shrank, and the account's owner count fell from 1 to 0. The engine counted any non-zero stored quality as an interest and kept the line. It now treats a stored QUALITY_ONE as no quality, as rippled does. The mainnet transaction is pinned as a regression vector. Fixed in e1140ef.
- **Finding F408 (AMM deposit: the minimum is not the amount issued).** At ledger 107,314,214 an account deposited 1.021192 XRP into an XRP/token AMM pool as a single-asset deposit, naming 56,639.2869999 LP tokens in the transaction's LPTokenOut field. In a single-asset deposit that field is only a minimum: rippled issues the LP tokens the deposited amount buys at the pool's price, and refuses the deposit (tecAMM_FAILED) only when that is fewer than the minimum. The XRP bought 57,211.40098 tokens, and mainnet issued 57,211.40098. The engine moved the same XRP but issued exactly the minimum, about 572 tokens short. It now issues what the deposit buys and refuses a deposit that falls below its minimum. The same code decided a second kind of single-asset deposit, in which the account names the tokens it wants: rippled issues that request rounded down to the precision at which the pool's total can record it, and the engine now does too (it issued the unrounded request). The mainnet transaction is pinned as a regression vector; the refusal and the rounding, which no mainnet transaction has exercised yet, are covered by unit tests. Fixed in 1fd1b9b.
- **The native-engine comparison lost step at 08:26 EDT, and was repaired.** Before verifying a ledger with libxrpl, the validator checks that its stored state is aligned with that ledger: the sender of the ledger's first sequence-numbered transaction must hold that sequence number. At ledger 107,314,806 that sender's account flags (it blocks incoming trust lines and NFT offers, flags 0x24000000) begin with the same byte that marks the sequence field, and the check searched for that byte instead of reading the fields in order, so it read 36 where the account held 62,037,183. It concluded the state was out of step, declined to verify the ledger, and handed the native-engine comparison an empty result; the comparison, which resets its in-memory copy of the ledger to the libxrpl result after every ledger, lost that ledger's changes and from then on reported a difference on every ledger: 2,303 ledgers, 107,314,806 to 107,317,138, until the deployment rebuilt its copy. None were engine errors and the validator was unaffected: its state hash matched the network on every ledger throughout. Of the flagged ledgers the automated re-check reached, 65 matched mainnet on every transaction (7,398 of 7,398); the other two, 107,314,819 and 107,314,838, match on the full ledger state with the flow engine the validator runs (105 of 105 and 251 of 251), the re-check having used the engine's older flow code for the first and an incomplete pre-state for one failed payment in the second. Fixed in 50a05ae: the check reads the account's fields in order, and a ledger the libxrpl side did not verify now makes the comparison rebuild its copy from the validator's stored state instead of adopting an empty result. A replay against mainnet of one ledger in twenty from the affected range, with the flow engine the validator runs, matched on every transaction: 117 ledgers, 16,302 of 16,302 transactions (finished 16:17 EDT).

- **Deployment.** Build 50a05ae (F407, F408 and the fix above) replaced the live build at 10:58 EDT, after its gate (1,697 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.6 seconds and resumed two ledgers behind the network. The native-engine comparison rebuilt its copy of the ledger (19,997,488 objects in 242 seconds) and has reported no difference since. The health check matched 411 ledgers with no mismatches, and soak #36 began at 11:25 EDT (seed ledger 107,317,138).
- **Soak #36.** At 18:18 EDT, 6.9 hours in, soak #36 of build 50a05ae was healthy: 6,801 state-hash matches and no mismatches, level with the network, on its primary reference node with no failovers, and no engine receipts. The native-engine comparison agreed byte for byte on all 6,795 ledgers it compared, and detected the one-drop canary it planted to prove the comparison is not blind.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Sponsor-paid NFTokenPage reserves for mint-to-player** (nft) This extends Sponsor (XLS-68) to cover `NFTokenPage`, a shared object that holds many tokens and that the current sponsorship model does not handle. A
- **Not-before ledger bound for pre-signed transactions** (hooks) Add a common optional field `sfFirstLedgerSequence` (UInt32) that mirrors `LastLedgerSequence`. It is covered by the signature. `preflight` rejects a
- **Directed Offers for RFQ block-trade settlement** (settlement) Add a flag `tfDirected` and the field `sfDestination` to `OfferCreate`. A directed offer never crosses on creation and is never put in a `BookDirector

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-29 08:53	09-29 09:06	xrpl-validator	`e1140ef`	fix(xrpl-ledger): F407 — a stored QualityIn of QUALITY_ONE is no interest; the default line is deleted
09-29 12:16	09-29 12:16	xrpl-validator	`1fd1b9b`	fix(xrpl-ledger): F408 — an AMM deposit mints the tokens its mode computes; LPTokenOut is never the mint
09-29 13:37	09-29 13:42	xrpl-validator	`50a05ae`	fix(xrpl-node): the era sentinel walks the AccountRoot's fields; an unverified ledger re-hydrates the native shadow

Times in UTC. A commit appears on the day it was committed to main or merged into it.